

ECON 4720: HA03

Rahul Sunilkumar

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Problem 1

Are the following vectors linearly independent?

1a

$$a_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 7 \\ 6 \end{pmatrix}, \quad a_3 = \begin{pmatrix} 12 \\ 10 \end{pmatrix}$$

Solution Attempt These vectors live in $\mathbb{R}^2$, so any set of three vectors must be linearly dependent. To make this explicit, solve

$$c_1 a_1 + c_2 a_2 + c_3 a_3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \iff \begin{cases} 3c_1 + 7c_2 + 12c_3 = 0, \\ 2c_1 + 6c_2 + 10c_3 = 0. \end{cases}$$

Row-reducing (work condensed) gives

$$\begin{bmatrix} 3 & 7 & 12 \\ 2 & 6 & 10 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{3}{2} \end{bmatrix},$$

so

$$c_1 = -\frac{1}{2} c_3, \quad c_2 = -\frac{3}{2} c_3,$$

with c_3 free. Taking $c_3 = 1$ yields the concrete dependence

$$-\frac{1}{2} a_1 - \frac{3}{2} a_2 + 1 \cdot a_3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

and, clearing denominators,

$$-1 \cdot a_1 - 3 \cdot a_2 + 2 \cdot a_3 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Therefore, the vectors are linearly dependent (not independent).

□

1b

$$a_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad a_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Solution Attempt Test

$$c_1 a_1 + c_2 a_2 + c_3 a_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Writing components,

$$c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Thus

$$c_1 = 0, \quad c_2 = 0, \quad c_3 = 0,$$

so the only solution is the trivial one.

Therefore, the vectors are linearly independent.

□

Problem 2

Find determinants of the following matrices:

$$A = \begin{bmatrix} 1 & 5 & 7 \\ 3 & 2 & 8 \\ 6 & 1 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 3 & 2 \\ 6 & -2 & 3 \\ 7 & 10 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 13 & 4 & 1 \\ 0 & 4 & 1 \\ -7 & 2 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} -4 & 4 & 12 \\ 3 & -3 & -9 \\ 8 & 2 & 6 \end{bmatrix}$$

Solution Attempt Cofactor along row 1:

$$\det A = 1 \begin{vmatrix} 2 & 8 \\ 1 & 9 \end{vmatrix} - 5 \begin{vmatrix} 3 & 8 \\ 6 & 9 \end{vmatrix} + 7 \begin{vmatrix} 3 & 2 \\ 6 & 1 \end{vmatrix} = 1(18 - 8) - 5(27 - 48) + 7(3 - 12) = 10 + 105 - 63 = 52.$$

Cofactor along row 1:

$$\det B = -1 \begin{vmatrix} -2 & 3 \\ 10 & 0 \end{vmatrix} - 3 \begin{vmatrix} 6 & 3 \\ 7 & 0 \end{vmatrix} + 2 \begin{vmatrix} 6 & -2 \\ 7 & 10 \end{vmatrix} = -1(-30) - 3(-21) + 2(60+14) = 30+63+148 = 241.$$

Expand down column 1 (uses the zero):

$$\det C = 13 \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix} + 0 \cdot (\dots) + (-7) \begin{vmatrix} 4 & 1 \\ 4 & 1 \end{vmatrix} = 13(12-2) + (-7)(0) = 130.$$

Note $R_2 = (-\frac{3}{4})R_1$ (rows dependent), so

$$\det D = 0.$$

$\det A = 52, \quad \det B = 241, \quad \det C = 130, \quad \det D = 0.$
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□

Problem 3

Using definitions, check whether the following matrices are positive definite or positive semidefinite:

Problem 3a

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}, \quad E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Solution Attempt Using the definition: a matrix M is PSD if $x^\top Mx \geq 0$ for all $x \neq 0$, and PD if $x^\top Mx > 0$ for all $x \neq 0$.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$x^\top Ax = x_1^2 + x_2^2 > 0 \text{ for all } x \neq 0.$$

$\Rightarrow$ **PD.**

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$x^\top Bx = x_1^2 \geq 0, \quad \text{and } x^\top Bx = 0 \text{ for } x = (0, 1)^\top \neq 0.$$

$\Rightarrow$ **PSD, not PD.**

$$C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$x^\top Cx = 2x_1x_2.$$

Takes both signs (e.g., $x = (1, -1)^\top \Rightarrow -2$ and $x = (1, 1)^\top \Rightarrow 2$).

$\Rightarrow$ **indefinite** (neither PSD nor PD).

$$D = \begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$$

$$x^\top Dx = x_1^2 - 3x_1x_2 + x_2^2 = \left(x_1 - \frac{3}{2}x_2\right)^2 - \frac{5}{4}x_2^2.$$

Negative for $x = (1, 2)^\top$ ($1 - 6 + 4 = -1$), positive for $x = (1, 0)^\top$ ($= 1$).

$\Rightarrow$ **indefinite.**

$$E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$x^\top Ex = x_1x_2,$$

which takes both signs (e.g., $x = (1, -1)^\top \Rightarrow -1$, $x = (1, 1)^\top \Rightarrow 1$).

$\Rightarrow$ **indefinite.**

$A : \text{PD}; \quad B : \text{PSD (not PD)}; \quad C, D, E : \text{indefinite.}$
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□

Problem 3b

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

Solution Attempt Using the definition: M is PSD if $x^\top Mx \geq 0$ for all $x \neq 0$, and PD if $x^\top Mx > 0$ for all $x \neq 0$.

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ (symmetric)}$$

$$x^\top Ax = 2x_1x_3 + x_2^2.$$

Takes both signs (e.g., $x = (1, 0, -1)^\top \Rightarrow -2$, $x = (1, 0, 1)^\top \Rightarrow 2$).

$\Rightarrow$ indefinite (neither PSD nor PD).

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 0 \end{bmatrix} \text{ (symmetric)}$$

Test by quadratic form:

$$x = (1, -2, 1)^\top \Rightarrow x^\top Bx = -9 < 0, \quad x = (1, 1, 1)^\top \Rightarrow x^\top Bx = 27 > 0.$$

So it attains both signs.

$\Rightarrow$ indefinite (neither PSD nor PD).

$$C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \text{ (symmetric)}$$

$$x^\top Cx = x_1^2 + x_2^2 + x_3^2 + 2x_1x_3 = (x_1 + x_3)^2 + x_2^2 \geq 0,$$

and equals 0 for $x = (1, 0, -1)^\top \neq 0$.

$\Rightarrow$ PSD (not PD).

$$D = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix} \text{ (symmetric)}$$

Observe

$$D = -vv^\top, \quad v = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}.$$

Hence

$$x^\top Dx = -(v^\top x)^2 \leq 0, \text{ with } x = v \Rightarrow x^\top Dx < 0.$$

So it is negative semidefinite (nonpositive for all x).

$\Rightarrow$ not PSD/PD in the positive sense (NSD).

$A : \text{indefinite}, \quad B : \text{indefinite}, \quad C : \text{PSD (not PD)}, \quad D : \text{negative semidefinite (not PSD/PD)}.$

□

Problem 4

Find the rank of each the following matrices:

$$A = \begin{bmatrix} 3 & 6 & 4 & 8 \\ 2 & 7 & 1 & 9 \\ 4 & 2 & 5 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 18 & 2 \\ 7 & -4 \\ 6 & 11 \end{bmatrix}, \quad C = \begin{bmatrix} -5 & 8 & 11 & 0 \\ 13 & 3 & 0 & 2 \\ 10 & 0 & -6 & 2 \end{bmatrix}$$

Solution Attempt Find the rank of each matrix.

$$A = \begin{bmatrix} 3 & 6 & 4 & 8 \\ 2 & 7 & 1 & 9 \\ 4 & 2 & 5 & 0 \end{bmatrix}$$

Condensed row-reduction:

$$\begin{bmatrix} 3 & 6 & 4 & 8 \\ 2 & 7 & 1 & 9 \\ 4 & 2 & 5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & \frac{4}{3} & \frac{8}{3} \\ 0 & 1 & -\frac{5}{3} & \frac{7}{3} \\ 0 & -6 & -\frac{2}{3} & -\frac{32}{3} \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & \frac{19}{11} \\ 0 & 0 & 1 & \frac{10}{11} \end{bmatrix}.$$

Three pivots $\Rightarrow \text{rank}(A) = 3$.

$$(Equivalently, \det \text{ of the } 3 \times 3 \text{ left block } \begin{vmatrix} 3 & 6 & 4 \\ 2 & 7 & 1 \\ 4 & 2 & 5 \end{vmatrix} = -33 \neq 0.)$$

$$B = \begin{bmatrix} 18 & 2 \\ 7 & -4 \\ 6 & 11 \end{bmatrix}$$

Condensed row-reduction:

$$\begin{bmatrix} 18 & 2 \\ 7 & -4 \\ 6 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Two pivots $\Rightarrow \text{rank}(B) = 2$.

$$(Equivalently, the } 2 \times 2 \text{ minor } \begin{vmatrix} 18 & 2 \\ 7 & -4 \end{vmatrix} = -86 \neq 0.)$$

$$C = \begin{bmatrix} -5 & 8 & 11 & 0 \\ 13 & 3 & 0 & 2 \\ 10 & 0 & -6 & 2 \end{bmatrix}$$

Condensed row-reduction:

$$\begin{bmatrix} -5 & 8 & 11 & 0 \\ 13 & 3 & 0 & 2 \\ 10 & 0 & -6 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & \frac{5}{64} \\ 0 & 1 & 0 & \frac{21}{64} \\ 0 & 0 & 1 & -\frac{13}{64} \end{bmatrix}.$$

Three pivots $\Rightarrow \text{rank}(C) = 3$.

(Equivalently, $\det \begin{bmatrix} -5 & 8 & 11 \\ 13 & 3 & 0 \\ 10 & 0 & -6 \end{bmatrix} = 384 \neq 0$.)

$\text{rank}(A) = 3, \quad \text{rank}(B) = 2, \quad \text{rank}(C) = 3.$

□

Problem 5

Find such value of the parameter α that the following vectors are linearly dependent:

$$a_1 = \begin{pmatrix} 2 \\ \alpha \\ -2 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}, \quad a_3 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$$

Solution Attempt The vectors are linearly dependent if the determinant of the matrix formed by placing them as columns equals 0:

$$\det \begin{bmatrix} 2 & 3 & -1 \\ \alpha & 0 & 2 \\ -2 & 4 & 1 \end{bmatrix} = 0.$$

Compute the determinant by expansion along the first row:

$$2 \begin{vmatrix} 0 & 2 \\ 4 & 1 \end{vmatrix} - 3 \begin{vmatrix} \alpha & 2 \\ -2 & 1 \end{vmatrix} + (-1) \begin{vmatrix} \alpha & 0 \\ -2 & 4 \end{vmatrix} = 0.$$

Simplify each minor:

$$2(0 \cdot 1 - 8) - 3(\alpha \cdot 1 - (-4)) - 1(\alpha \cdot 4 - 0) = 0,$$

$$2(-8) - 3(\alpha + 4) - (4\alpha) = 0,$$

$$-16 - 3\alpha - 12 - 4\alpha = 0,$$

$$-7\alpha - 28 = 0.$$

Hence,

$$\alpha = -4.$$

$$\boxed{\alpha = -4}$$

□

Problem 6

Solve the following systems of linear questions:

Problem 6a

$$\begin{cases} x + y + z = 11 \\ x - y - z = -3 \\ -x + y - z = 5 \\ 3x - y + 2z = 2 \end{cases}$$

Solution Attempt Add the first two equations to eliminate y, z :

$$(x + y + z) + (x - y - z) = 11 + (-3) \Rightarrow 2x = 8 \Rightarrow x = 4.$$

Use $x = 4$ in the 2nd and 3rd equations:

$$4 - y - z = -3 \Rightarrow y + z = 7, \quad -4 + y - z = 5 \Rightarrow y - z = 9.$$

Solve the 2×2 system:

$$y = \frac{(7+9)}{2} = 8, \quad z = 7 - y = -1.$$

Check in the 4th equation: $3(4) - 8 + 2(-1) = 12 - 8 - 2 = 2.$

$$\boxed{(x, y, z) = (4, 8, -1).}$$

□

Problem 6b

$$\begin{cases} x + y + z + t + p = 17 \\ x - y - z - t - p = -5 \\ z + t + p + y = 11 \\ p - x - y = 1 \\ -t + x = 10 \end{cases}$$

Solution Attempt ## Problem 6b

$$\begin{cases} x + y + z + t + p = 17, \\ x - y - z - t - p = -5, \\ y + z + t + p = 11, \\ p - x - y = 1, \\ -t + x = 10. \end{cases}$$

Solution Attempt

From the last two equations:

$$t = x - 10, \quad p = 1 + x + y.$$

Use these in $y + z + t + p = 11$:

$$y + z + (x - 10) + (1 + x + y) = 11 \Rightarrow 2x + 2y + z = 20 \Rightarrow z = 20 - 2x - 2y.$$

Now plug t, p, z into $x + y + z + t + p = 17$:

$$x + y + (20 - 2x - 2y) + (x - 10) + (1 + x + y) = 17 \Rightarrow 3x + 2y + z = 26 \Rightarrow x = 6.$$

Hence

$$t = -4, \quad p = 7 + y, \quad z = 8 - 2y.$$

The second equation is then automatically satisfied, so y is free.

$$(x, y, z, t, p) = (6, y, 8 - 2y, -4, 7 + y), \quad y \in \mathbb{R}.$$

□

Problem 6c

$$\begin{cases} x + y + z + t = -6 \\ x - y - z - t = 20 \\ y - x = -39 \\ 2x + 3t + y - 4z = 32 \end{cases}$$

Solution Attempt From $y - x = -39$ we get

$$y = x - 39.$$

Substitute into the first two equations:

$$x + (x - 39) + z + t = -6 \Rightarrow 2x + z + t = 33, x - (x - 39) - z - t = 20 \Rightarrow z + t = 19.$$

Combine to solve for x :

$$2x + (z + t) = 33 \Rightarrow 2x + 19 = 33 \Rightarrow x = 7, \quad y = x - 39 = -32.$$

Use $z + t = 19$ and the last equation:

$$2x + 3t + y - 4z = 32 \Rightarrow 14 + 3t - 32 - 4z = 32 \Rightarrow 3t - 4z = 50.$$

With $t = 19 - z$:

$$3(19 - z) - 4z = 50 \Rightarrow 57 - 7z = 50 \Rightarrow z = 1, \quad t = 18.$$

$$(x, y, z, t) = (7, -32, 1, 18).$$

□

Problem 6d

$$\begin{cases} -4x - 5y - 5z = -74 \\ -3x - 5y - z = -4 \\ -x + 7y + 6z = 48 \end{cases}$$

From the first two equations, eliminate y by subtraction:

$$(-4x - 5y - 5z) - (-3x - 5y - z) = -74 - (-4) \Rightarrow -x - 4z = -70 \Rightarrow x = 70 - 4z.$$

Use the third equation to relate y and z :

$$-(70 - 4z) + 7y + 6z = 48 \Rightarrow 7y + 10z = 118 \Rightarrow y = \frac{118 - 10z}{7}.$$

Substitute x and y into $-3x - 5y - z = -4$:

$$-3(70 - 4z) - 5\left(\frac{118 - 10z}{7}\right) - z = -4.$$

Multiply by 7 to clear denominators and solve for z (work condensed):

$$-1302 + 129z = -28 \Rightarrow z = 16.$$

Then

$$x = 70 - 4 \cdot 16 = 6, \quad y = \frac{118 - 10 \cdot 16}{7} = -6.$$

$$(x, y, z) = (6, -6, 16).$$

□

Problem 6e

$$\begin{cases} x + 2y - z + 3t = 0 \\ x + 2y + 2z - t = 1 \\ 3x + 6y + 5t = 1 \\ 4x + 8y - z + 8t = 1 \end{cases}$$

Solution Attempt Subtract the first equation from the second:

$$(x + 2y + 2z - t) - (x + 2y - z + 3t) = 1 - 0 \Rightarrow 3z - 4t = 1 \Rightarrow z = \frac{1 + 4t}{3}.$$

Use this in the first equation to get x :

$$x + 2y - \frac{1 + 4t}{3} + 3t = 0 \Rightarrow x = -2y + \frac{1 - 5t}{3}.$$

With these x, z , the third and fourth equations are automatically satisfied (they reduce to identities), so y and t are free parameters.

Therefore the solution set is

$$(x, y, z, t) = \left(\frac{1 - 5t}{3} - 2y, y, \frac{1 + 4t}{3}, t \right), \quad y, t \in \mathbb{R}.$$

Equivalently, in vector form,

$$\begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \\ 0 \\ \frac{1}{3} \\ 0 \end{pmatrix} + y \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -\frac{5}{3} \\ 0 \\ \frac{4}{3} \\ 1 \end{pmatrix}.$$

Infinitely many solutions with y, t free.

□

Problem 7

For which values of λ the following matrix is invertible?

$$A = \begin{bmatrix} 1 & -2 & 2 \\ \lambda & 3 & 0 \\ 2 & 1 & 1 \end{bmatrix}$$

Solution Attempt A is invertible iff $\det A \neq 0$. Compute

$$\det A = 1 \begin{vmatrix} 3 & 0 \\ 1 & 1 \end{vmatrix} - (-2) \begin{vmatrix} \lambda & 0 \\ 2 & 1 \end{vmatrix} + 2 \begin{vmatrix} \lambda & 3 \\ 2 & 1 \end{vmatrix} = 3 + 2\lambda + 2(\lambda - 6) = 4\lambda - 9.$$

A is invertible iff $4\lambda - 9 \neq 0 \iff \lambda \neq \frac{9}{4}$.

□

Problem 8

Find all values a , for which the system of linear equations does not have any solution.

$$\begin{cases} 2x + (9a^2 - 2)y = 3a \\ x + y = 1 \end{cases}$$

Solution Attempt The coefficient matrix is

$$M = \begin{bmatrix} 2 & 9a^2 - 2 \\ 1 & 1 \end{bmatrix}, \quad \det M = 2 \cdot 1 - (9a^2 - 2) \cdot 1 = 4 - 9a^2.$$

If $\det M \neq 0$ (i.e., $a \neq \pm \frac{2}{3}$), there is a unique solution. If $\det M = 0$ (i.e., $a = \pm \frac{2}{3}$), check consistency:

For $a = \frac{2}{3}$:

$$2x + (4 - 2)y = 2 \iff 2x + 2y = 2, \quad \text{and} \quad x + y = 1.$$

These are the same equation $\Rightarrow$ infinitely many solutions (consistent).

For $a = -\frac{2}{3}$:

$$2x + 2y = -2, \quad \text{but} \quad x + y = 1 \Rightarrow 2x + 2y = 2,$$

which is a contradiction $\Rightarrow$ no solution.

No solution exactly when $a = -\frac{2}{3}$.

□

Problem 9

Find a value of b such that system of linear equations has infinitely many solutions:

$$\begin{cases} 6x + 2by = 3 \\ 4x + 2y = 2 \end{cases}$$

Solution Attempt A system of two linear equations in two variables has infinitely many solutions when one equation is a scalar multiple of the other.

That is, we must have proportional coefficients:

$$\frac{6}{4} = \frac{2b}{2} = \frac{3}{2}.$$

Simplify:

$$\frac{6}{4} = \frac{3}{2}, \quad \frac{2b}{2} = b, \quad \frac{3}{2} = \frac{3}{2}.$$

So we require $b = \frac{3}{2}$.

$$b = \frac{3}{2}.$$

□

Problem 10

For the following matrices check the statement: “If a matrix X has full rank then the matrix $X'X$ is invertible” and find the inverse $(X'X)^{-1}$:

Problem 10a

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Solution Attempt If X has full column rank, then $X'X$ is invertible.

Here $X \in \mathbb{R}^{4 \times 2}$ and $\text{rank}(X) = 2$ (its two columns are the standard basis vectors in $\mathbb{R}^4$), so X has full column rank. Compute

$$X'X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2,$$

which is invertible.

$$(X'X)^{-1} = I_2.$$

□

Problem 10b

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$$

Solution Attempt Columns:

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

They are linearly independent, so $\text{rank}(X) = 2$ (full column rank) and thus $X'X$ is invertible.

Compute

$$X'X = \begin{bmatrix} x_1^\top x_1 & x_1^\top x_2 \\ x_2^\top x_1 & x_2^\top x_2 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}, \quad \det(X'X) = 4 \cdot 1 - 1 \cdot 1 = 3 \neq 0.$$

Hence

$$(X'X)^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}.$$

$$X'X = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}, \quad (X'X)^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}.$$

□

Problem 10c

$$X = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 0 \\ 4 & 1 \end{bmatrix}$$

Solution Attempt Columns are independent (clearly not multiples), so $\text{rank}(X) = 2$ (full column rank) and $X'X$ is invertible. Compute

$$X'X = \begin{bmatrix} x_1^\top x_1 & x_1^\top x_2 \\ x_2^\top x_1 & x_2^\top x_2 \end{bmatrix} = \begin{bmatrix} 1^2 + 2^2 + 3^2 + 4^2 & 1 \cdot 1 + 2 \cdot 3 + 3 \cdot 0 + 4 \cdot 1 \\ 1^2 + 3^2 + 0^2 + 1^2 & 11 \end{bmatrix} = \begin{bmatrix} 30 & 11 \\ 11 & 11 \end{bmatrix}.$$

Its determinant is $\det(X'X) = 30 \cdot 11 - 11 \cdot 11 = 209 \neq 0$, so

$$(X'X)^{-1} = \frac{1}{209} \begin{bmatrix} 11 & -11 \\ -11 & 30 \end{bmatrix}.$$

$$X'X = \begin{bmatrix} 30 & 11 \\ 11 & 11 \end{bmatrix}, \quad (X'X)^{-1} = \frac{1}{209} \begin{bmatrix} 11 & -11 \\ -11 & 30 \end{bmatrix}.$$

□